

Hospitality Analytics Dashboard

Project Report

Executive Summary

This report documents a **Streamlit-based Hospitality Analytics Dashboard** that analyzes hotel booking data, performs data preprocessing, generates interactive visualizations, and predicts booking status using a **Random Forest Classifier**. The dashboard enables hotel management to monitor business performance, booking trends, revenue generation, and customer behavior through an interactive interface.

Problem Statement

Hospitality businesses generate large volumes of booking data from multiple properties and booking platforms. Manual analysis makes it difficult to identify revenue trends, booking patterns, customer preferences, and operational performance.

This project transforms raw hotel booking data into meaningful insights and predictive analytics, enabling management to make data-driven decisions.

Objectives

- Analyze hotel booking performance.
- Generate business KPIs.
- Visualize revenue across cities and properties.
- Analyze booking status and customer ratings.
- Evaluate booking platform performance.
- Predict booking status using Machine Learning.
- Build an interactive Streamlit dashboard for business users.

Dataset

The project uses the **Hospitality_Excel.xlsx** dataset (Final_Data sheet).

The dataset contains **134,590 booking records** with hotel booking information.

Important Fields

- Booking ID
 - Property ID
 - Booking Date
 - Check-in Date
 - Checkout Date
 - Number of Guests
 - Room Category
 - Room Class
 - Booking Platform
 - Booking Status
 - Revenue Generated
 - Revenue Realized
 - Property Name
 - Category
 - City
 - Ratings Given
 - Month
 - Day Type
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Data Preprocessing (ETL)

The ETL process included:

- Loading the Final_Data sheet
- Handling missing values
- Removing duplicate records
- Converting booking dates into datetime format
- Creating engineered features:
 - Stay Duration
 - Revenue Per Guest
 - Booking Month
 - Booking Year
 - Booking Day
- Exporting the cleaned dataset for dashboard development

Exploratory Data Analysis

The following analyses were performed:

- Revenue by City
 - Revenue by Property
 - Revenue by Booking Platform
 - Booking Status Distribution
 - Room Class Analysis
 - Ratings Distribution
 - Revenue Trend Analysis
 - Average Stay Duration
 - Property Performance Comparison
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Machine Learning Model

Algorithm

Random Forest Classifier

Input Features

- City
- Room Class
- Booking Platform
- Number of Guests
- Month
- Ratings Given

Target Variable

Booking Status

Model Workflow

Clean Dataset

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Feature Selection

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Encoding Categorical Variables

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Train-Test Split

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Random Forest Training

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Model Evaluation

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Booking Status Prediction

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Prediction displayed in Streamlit Dashboard

Streamlit Dashboard Modules



Home

- Project Overview
- KPI Cards
- Dataset Preview
- Summary Statistics



Executive Overview

- Revenue KPIs
- Revenue by City
- Booking Status Distribution
- Booking Platform Analysis
- Ratings Distribution



Property Performance

- Revenue by Property
- Room Class Performance
- Property Summary
- Interactive Filters



Revenue Analysis

- Revenue Trend
- Revenue by City
- Revenue by Property
- Revenue by Room Class
- Revenue by Booking Platform



Prediction

- User Input Form
- Booking Status Prediction
- Machine Learning Model Information

Key Business Insights

1. Revenue varies significantly across different cities and hotel properties.
2. Certain booking platforms contribute more revenue than others.
3. Customer ratings help identify high-performing properties.
4. Room class directly influences revenue generation.
5. Stay duration impacts revenue per guest.
6. Predictive analytics can estimate booking outcomes before check-in.

Recommendations

- Improve marketing efforts for low-performing properties.
 - Increase promotion through high-performing booking platforms.
 - Focus on improving customer ratings and service quality.
 - Optimize room allocation based on demand.
 - Use predictive analytics to reduce cancellations and improve operational planning.
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Tools & Technologies

Programming

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- Scikit-learn
- Joblib

Dashboard

- Streamlit

Business Intelligence

- Power BI

IDE

- Visual Studio Code

Dataset

- Microsoft Excel
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Conclusion

The Hospitality Analytics Dashboard successfully demonstrates an end-to-end data analytics workflow, including ETL, exploratory data analysis, machine learning, and interactive dashboard development. The solution enables hotel management to monitor key business metrics, understand booking behavior, evaluate revenue performance, and predict booking status using machine learning. This project showcases practical data analytics and business intelligence skills applicable to real-world hospitality operations.